HACK-ADEMY

Tools for Hackathon Success

entelect
everything is possible

Content

- 🚀 Introduction
- ⚙️ Understanding Optimisation
- 🤖 AI Strategies & Tips
- 🌐 Website Walkthrough
- 🦏 Hands-on Prac

What is Entelect Hackathons?

The short version:

Timed boxed coding competition

- One problem statement, endless possible solutions.
- Any programming language is welcome
- AI usage is embraced

Everything runs through the hackathon website:

- Download the problem statement and inputs.
- Upload solution.
- Automatic scoring, in seconds.
- Live leaderboard.

Where you will find them:

- Community hackathons – HackIT.
- Community workshops and events.
- University Cups.

Optimisation Problems

Fundamental computational challenges used to teach and evaluate algorithm design. The best solution depends on constraints such as time, memory, and input size.

Main Categories

Dynamic Programming (DP)

- Solves overlapping subproblems by storing intermediate results.
- Examples: Knapsack, Coin Change, Longest Common Subsequence (LCS).

Greedy Algorithms

- Makes the best local choice at each step to achieve an optimal overall solution.
- Examples: Activity Selection, Dijkstra's Shortest Path, Huffman Encoding.

Graph Algorithms & Searching:

- Explores networks and relationships for navigation and optimisation.
- Examples: Travelling Salesman Problem (TSP), Depth-First Search (DFS), Breadth-First Search (BFS).

Optimisation in Day to Day

Driving from Sandton to O. R. Tambo International Airport.

Alternative Routes

Multiple routes are calculated simultaneously.

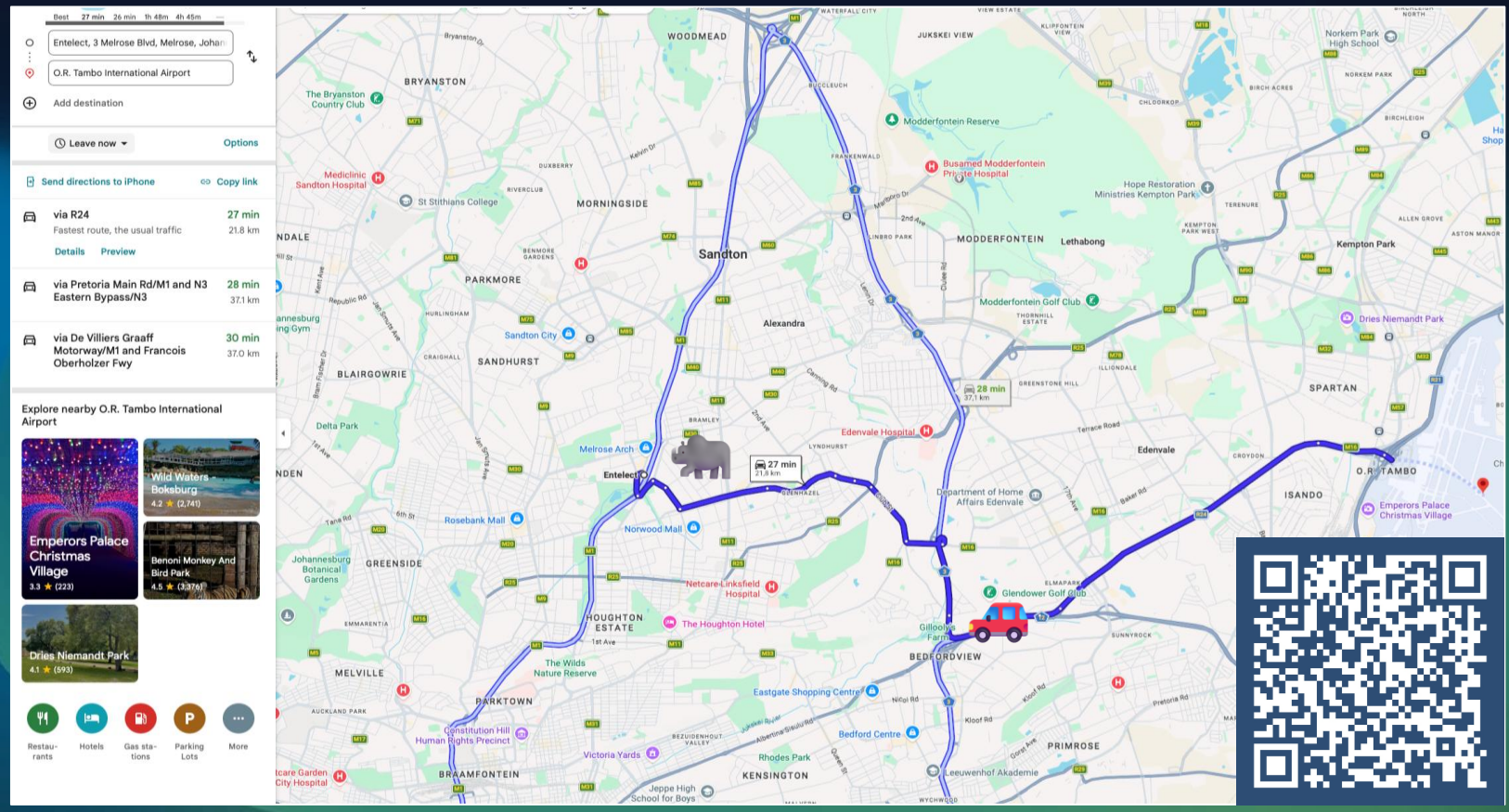
Real-Time Updates

Google Maps continuously receives live traffic information.

Route selection

Shortest path does not always mean fastest.

Highways vs local roads.



Strategies

Two approaches:

Know where to start

- Sketch out your idea.
- Use AI as a reviewer rather than a solution generator.

Unsure of where to start

- Summarise problem.
- Explain inputs & outputs.
- Identify constraints.
- Explain scoring.

Divide and Conquer — only AFTER everyone is on the same page

- Everyone needs to understand the problem.
- Brainstorm approaches and agree on boilerplate code.
- Then divide and conquer.

AI Tips

Think alongside the model:

- Understand every Solution.
- Ask the model to generate documentation.
- Review generated code.
- Try to improve upon the AI's output.

Iterate with Models and Solutions:

- Ask one LLM to help generate and refine the prompt for another.
- Feed the solution plan of one LLM to another model for revision.
- Try to improve upon the AI's output.
- Generate a few solutions and improve upon the best one.

Prompt Well:

- Use structured markdown.
- Be detailed and provide all files and context.
- Ask the LLM to tell you if information is missing or if something is unclear.

Pitfalls:

- Blindly trusting generated code.
- Not verifying assumptions.
- Spending hours prompt-engineering instead of testing.

The Problem Statement

Objective

What are you actually solving?

What does “done” look like?

Summarise it in one sentence before you code.

Inputs

What data do I get?

What format is it in?

Ask: *how will I load this into my code?*

Constraints

The rules of the game.

Limits, formats, and fine print.

Where edge cases hide.

Break them – your submission is invalid.

Outputs

What exactly do you submit?

Match the required format precisely.

Scoring

How are points calculated?

Optimise for the formula

Hints

Examples and guidance.

Take it level by level

Practice Problem: Ranger's Rescue Route

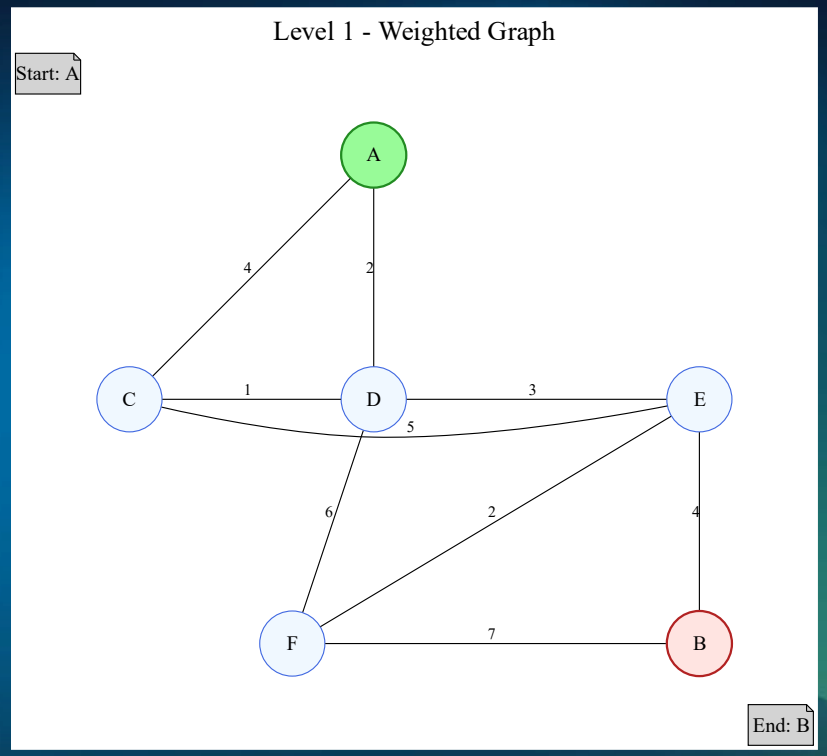
The mission

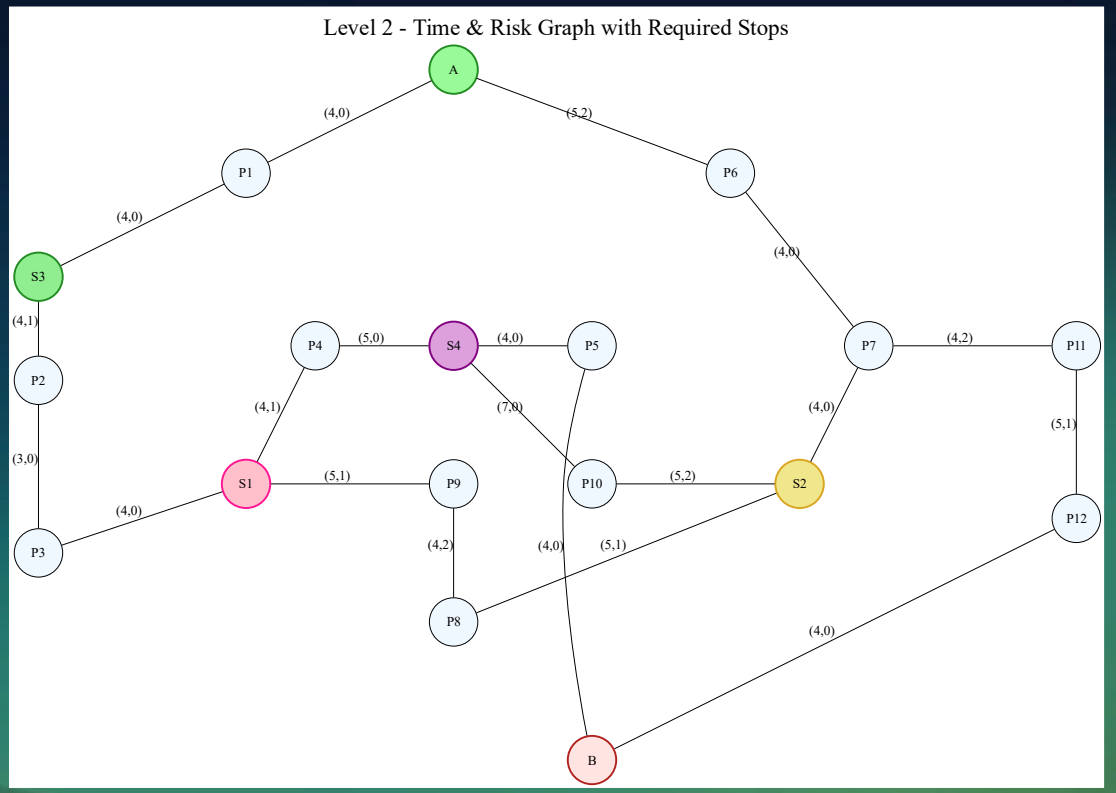
- You are a wildlife ranger. Find the cheapest route through a network of trails to reach a rescue.
- Objective:
  - Level 1: Quickest travel time from A to B
  - Level 2: Quickest travel with the least a mount of risk from point A to B while making 4 sperate estops to pick up some resources
- Input: the graph as an adjacency list – undirected, weighted.
- Output: { "route": ["A", "D", "E", "B"] } – node names in order.

Scoring

- Each level is a score out of 100

Practice Problem: Ranger's Rescue Route





Website Walkthrough
Using the Website

- Everything lives in one place: the problem statement, level inputs, submissions, and the leaderboard.
- Log in, register for a hackathon, and download the problem statement and level files on the day.
- Read first, code later: know the objective, format, and scoring before you write a line.
- When you are ready, head to the submissions page.
- Let's click through it together!

Website Walkthrough Submitting & Getting a Score

Upload two things:

- Your answer file (.txt or .json)

```
{
    "route": ["A", "D", "E", "B"]
}.
```

- Your source code (.zip).
  - Include a README in case a reviewer has to run your code.

Your score comes back instantly: optimal cost ÷ your cost × 100.

Website Walkthrough
Iterate, Iterate, Iterate

Your first submission is a baseline, not your final answer.

A perfect route on paper scores zero if it never gets submitted.

The loop:

- Submit early. Even a naive route earns points.
- Measure. Compare your score against the maximum (when available).
- Improve one thing, submit again, and watch the number climb.

Resources

Scan the QR code:

- Prompt guide.
- Problem statement
- Solution guide (pending)
- Slides

Happy Hacking!

Scan Me